Akoizo Original Practice - 2027 Rotation Form 1 - Answer key

Division C | Protein Modeling | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Alpha-helix H bonds mainly involve…

Unit: Structural levels and denaturation.
Worked-case data: Heating disrupts activity but the amino-acid sequence remains intact. What changed?

A. Only neighboring side-chain colors

B. Backbone groups

C. Only sulfur atoms

D. Only DNA bases

Correct: B

2. (2 points) Attachment and fusion are…

Unit: Viruses, entry, and immune recognition.
Worked-case data: An antibody binds a viral protein in an assay. Does that prove neutralization?

A. Only DNA processes

B. Absent in all enveloped viruses

C. Different entry-related steps

D. Always identical

Correct: C

3. (2 points) Rotation changes…

Unit: Molecular viewing and reproducible JUDE/Jmol workflows.
Worked-case data: Plan a figure highlighting one chain's helix region.

A. Primary sequence

B. Chemical bonds automatically

C. PDB identifier

D. Viewpoint

Correct: D

4. (2 points) Quaternary structure requires…

Unit: Structural levels and denaturation.
Worked-case data: Heating disrupts activity but the amino-acid sequence remains intact. What changed?

A. No interfaces

B. Multiple polypeptide subunits

C. Exactly four atoms

D. Only one residue

Correct: B

5. (2 points) An epitope is…

Unit: Viruses, entry, and immune recognition.
Worked-case data: An antibody binds a viral protein in an assay. Does that prove neutralization?

A. Every viral genome

B. A unit of salinity

C. A recognized molecular feature

D. The entire immune system

Correct: C

6. (2 points) Color by structure shows…

Unit: Molecular viewing and reproducible JUDE/Jmol workflows.
Worked-case data: Plan a figure highlighting one chain’s helix region.

A. Measured free energy

B. Every atom’s charge

C. A diagnosis

D. A structural classification

Correct: D

7. (2 points) Denaturation always cleaves peptide bonds?

Unit: Structural levels and denaturation.
Worked-case data: Heating disrupts activity but the amino-acid sequence remains intact. What changed?

A. Only at neutral pH

B. No

C. Always

D. Only for helices

Correct: B

8. (2 points) Binding alone proves neutralization?

Unit: Viruses, entry, and immune recognition.
Worked-case data: An antibody binds a viral protein in an assay. Does that prove neutralization?

A. Only for any antibody

B. Only in a diagram

C. No

D. Always

Correct: C

9. (2 points) The built-in lab executes full Jmol?

Unit: Molecular viewing and reproducible JUDE/Jmol workflows.
Worked-case data: Plan a figure highlighting one chain’s helix region.

A. Yes

B. Only at maximum zoom

C. Only for HA

D. No

Correct: D

10. (4 points) Compare motif and domain.

Unit: Structural levels and denaturation.
Worked-case data: Heating disrupts activity but the amino-acid sequence remains intact. What changed?

answer: A motif is a recurring arrangement; a domain is a larger structural/functional unit. (4 points).

11. (2 points) The assigned interval runs from residue 153 through 252, inclusive. How many numbered residue positions are included?

answer: 100 (2 points).

Numerical target 100; absolute tolerance 0.2. Accepted units: , .

12. (2 points) At the specified scale of 2 cm per residue, what backbone material length corresponds to 100 residues before folding?

answer: 200 cm (2 points).

Numerical target 200; absolute tolerance 0.4. Accepted units: , cm.

13. (2 points) State the assigned 2027 structure identifier, chain and residue interval, and distinguish these identifiers from a claim that all residues are experimentally resolved.

answer: PDB 3UBE, chain A, residues 153-252. A numbered interval specifies the requested region; experimental missing residues must be checked in the structure record. (2 points).

14. (2 points) Explain the functional relevance of hemagglutinin binding to sialic-acid receptors in influenza entry.

answer: Binding helps the virus attach to susceptible host-cell receptors and initiates the entry pathway; binding specificity is related to infection biology. This is State/National molecular detail in the supplied rules. (2 points).

15. (2 points) Why does identifying an alpha helix require backbone geometry rather than simply finding a hydrophobic sequence stretch?

answer: Secondary structure depends on backbone conformation and hydrogen-bond organization; residue composition alone does not uniquely establish the actual geometry. (2 points).

16. (2 points) A synthetic structure worksheet places two oppositely charged side chains close together. Describe a possible stabilizing interaction and a qualification.

answer: An electrostatic/salt-bridge interaction may stabilize the arrangement, but protonation state, solvent, distance and geometry affect its strength. (2 points).

17. (2 points) Explain why displaying a ligand-binding pocket without identifying chain and residue numbering can mislead a model reviewer.

answer: Equivalent residue numbers can refer to different chains or numbering schemes; reproducible selection requires structure, chain, residue identifiers and consistent coordinate context. (2 points).

18. (2 points) What is the distinction between a displayed molecular contact and a proven functional interaction?

answer: A geometrical contact is structural evidence; functional significance requires appropriate chemical context and supporting evidence rather than proximity alone. (2 points).

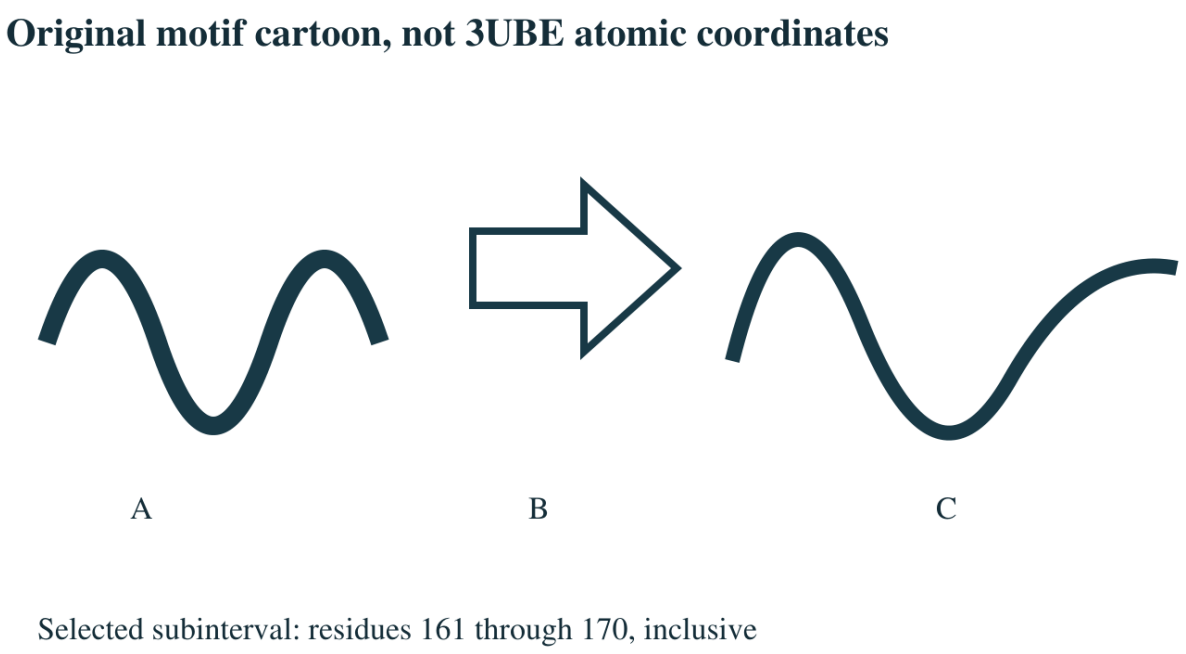
19. (2 points) A model includes a decorative feature unrelated to protein structure or function. Explain why it does not improve its scientific explanation.

answer: Useful features should communicate relevant structural or functional evidence and be identified in the legend; irrelevant decoration does not explain the molecule. (2 points).

20. (2 points) Explain why rotating a JUDE molecular view does not alter the primary sequence or chemical connectivity.

answer: Rotation changes the viewing transformation, not residue order or the underlying molecular bond structure. (2 points).

21. (2 points) Use Figure F1. Identify the secondary-structure motif convention represented by B.



F1. Backbone motifs and selected residue interval (original synthetic teaching figure)

answer: A beta strand, commonly drawn as an arrow in a beta sheet. (2 points).

22. (2 points) Use Figure F1. How many residue positions are included in the selected subinterval?

Refer to F1. Backbone motifs and selected residue interval (original synthetic teaching figure) shown with the first question in this figure group.

answer: 10 (2 points).

Numerical target 10; absolute tolerance 0.03. Accepted units: .

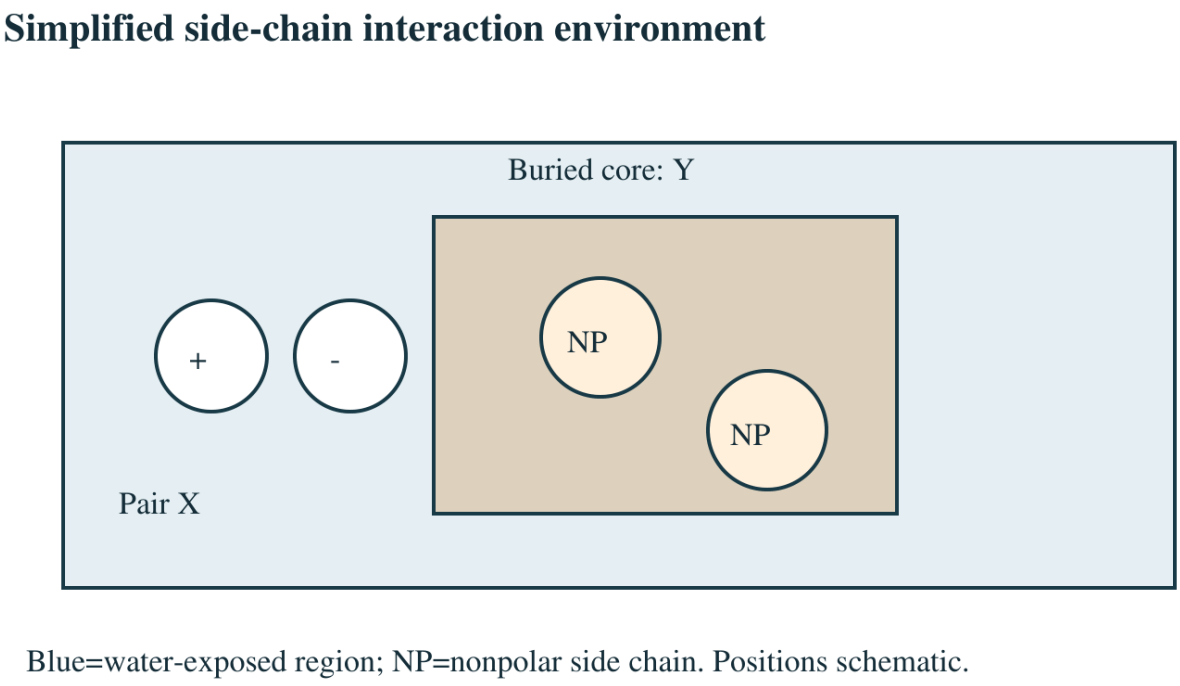
23. (2 points) Use Figure F1. At2 cm per residue, what model backbone length represents that selected subinterval?

Refer to F1. Backbone motifs and selected residue interval (original synthetic teaching figure) shown with the first question in this figure group.

answer: 20 cm (2 points).

Numerical target 20; absolute tolerance 0.06. Accepted units: , cm.

24. (2 points) Use Figure F2. What stabilizing interaction could occur at pair X? State one condition that affects it.



F2. Charge and burial in a folded protein (original synthetic teaching figure)

answer: An electrostatic/salt-bridge interaction; protonation state, distance, geometry and solvent can affect its contribution. (2 points).

25. (2 points) Use Figure F2. Explain why nonpolar side chains in Y can be favorable in an aqueous environment.

Refer to F2. Charge and burial in a folded protein (original synthetic teaching figure) shown with the first question in this figure group.

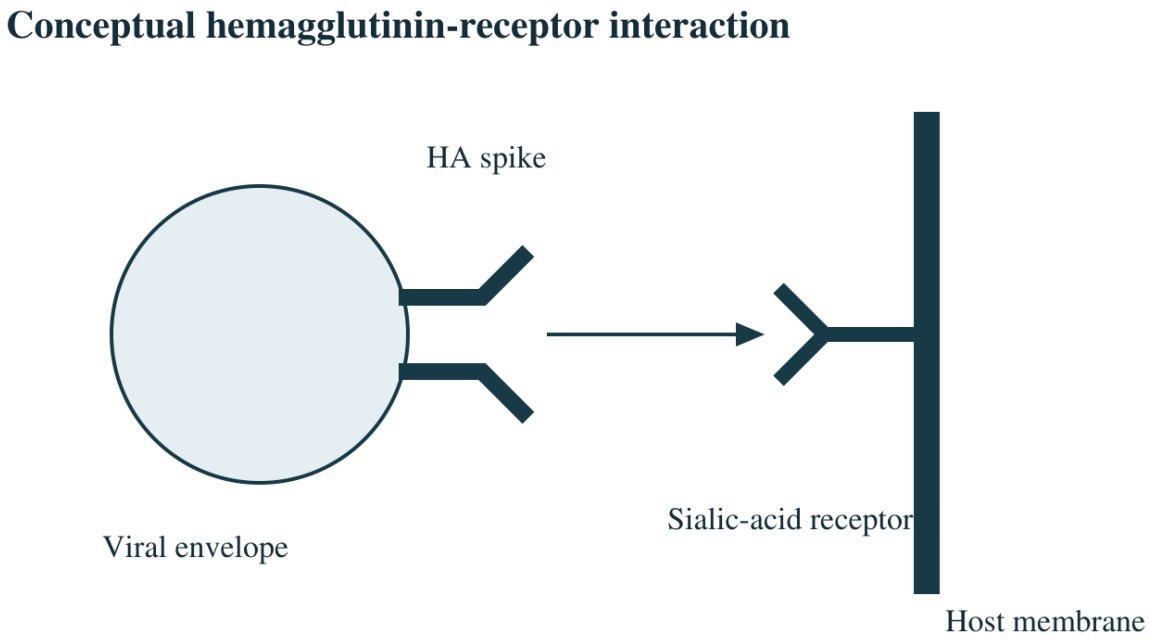
answer: Burying nonpolar side chains reduces their exposure to water and can contribute to hydrophobic stabilization of a fold. (2 points).

26. (2 points) Use Figure F2. Why could changing pH alter the displayed interaction at X?

Refer to F2. Charge and burial in a folded protein (original synthetic teaching figure) shown with the first question in this figure group.

answer: pH can change side-chain protonation and charge, altering electrostatic interactions and possibly stability. (2 points).

27. (2 points) Use Figure F3. [State/National] Which labeled viral feature directly participates in the depicted receptor attachment?



F3. Influenza attachment mechanism (original synthetic teaching figure)

answer: Hemagglutinin/HA spike. (2 points).

28. (2 points) Use Figure F3. [State/National] Predict how blocking the displayed receptor-binding interaction could affect this entry step.

Refer to F3. Influenza attachment mechanism (original synthetic teaching figure) shown with the first question in this figure group.

answer: It could reduce attachment through that HA-receptor interaction, while overall infectivity would require further experimental evidence and context. (2 points).

29. (2 points) Use Figure F3. [State/National] Why does this attachment drawing not prove that a bound particle has completed membrane fusion?

Refer to F3. Influenza attachment mechanism (original synthetic teaching figure) shown with the first question in this figure group.

answer: Attachment and fusion are distinct steps; the drawing shows binding geometry, not the subsequent conformational, membrane or functional evidence of fusion. (2 points).

30. (2 points) Use Figure F4. Which recorded field distinguishes the intended chain from other chains in the same structure?

Illustrative JUDE selection record; not a captured application screen

Field	Recorded selection
Structure	3UBE
Chain	A
Residue range	153-252 inclusive
Display	Backbone cartoon
View action	Rotate camera
Coordinate check	Check missing residues before building

F4. Molecular-view selection record (original synthetic teaching figure)

answer: Chain A. (2 points).

31. (2 points) Use Figure F4. Does the recorded camera rotation change primary sequence or bonds? Explain.

Refer to F4. Molecular-view selection record (original synthetic teaching figure) shown with the first question in this figure group.

answer: No. It changes the viewpoint rather than the underlying sequence or chemical connectivity. (2 points).

32. (2 points) Use Figure F4. Why is the final coordinate check important even when the numbered interval is correctly recorded?

Refer to F4. Molecular-view selection record (original synthetic teaching figure) shown with the first question in this figure group.

answer: Requested residue numbering does not guarantee every atom/residue is resolved in the experimental structure; missing segments must be verified and documented rather than invented. (2 points).